

Autonomous Flight and Navigation Architecture for Fixed-Wing UAVs

Mathematical Modeling, Transfer Functions, and Cascaded GNC Design

Autonomous Flight Control Systems Design

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Abstract

This report outlines the comprehensive Guidance, Navigation, and Control (GNC) architecture for a fully autonomous fixed-wing Unmanned Aerial Vehicle (UAV). To reduce mechanical complexity and simplify control allocation, the rudder has been eliminated; directional control is achieved via “Bank-to-Turn” maneuvers using ailerons. The architecture encompasses the linearized mathematical models, transfer functions, cascaded inner loops for attitude control, and outer loops for altitude, airspeed, and waypoint navigation, culminating in a state machine for automated flight phases.

1 Mathematical Model and Transfer Functions

The flight dynamics are decoupled into longitudinal and lateral-directional models. The mathematical representation assumes a linearized state-space formulation around a steady cruise flight condition.

1.1 Longitudinal Dynamics

The longitudinal model describes the forward velocity, vertical velocity, pitch rate, and pitch angle. The control inputs include the elevator ($\Delta\delta_e$) for attitude and engine thrust ($\Delta\delta_t$) for energy management.

State-Space Representation:

$$x_{lon} = \begin{bmatrix} \Delta u \\ \Delta w \\ q \\ \Delta\theta \end{bmatrix}, \quad u_{lon} = \begin{bmatrix} \Delta\delta_e \\ \Delta\delta_t \end{bmatrix} \quad (1)$$

$$\dot{x}_{lon} = A_{lon}x_{lon} + B_{lon}u_{lon} \quad (2)$$

$$A_{lon} = \begin{bmatrix} X_u & X_w & X_q & -g \cos \theta_0 \\ Z_u & Z_w + U_0 & -g \sin \theta_0 & 0 \\ M_u & M_w & M_q & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad B_{lon} = \begin{bmatrix} X_{\delta_e} & X_{\delta_t} \\ Z_{\delta_e} & Z_{\delta_t} \\ M_{\delta_e} & M_{\delta_t} \\ 0 & 0 \end{bmatrix} \quad (3)$$

Pitch Transfer Function: Considering only the elevator's effect on the pitch angle (ignoring speed variations for the short-period approximation), the transfer function contains a free integrator ($\frac{1}{s}$), indicating that direct angle control requires careful tuning or a cascaded approach:

$$\frac{\Theta(s)}{\Delta E(s)} = \frac{1}{s} \frac{M_{\delta_e} s + \left(\frac{M_\alpha Z_{\delta_e}}{V_0} - \frac{M_{\delta_e} Z_\alpha}{V_0} \right)}{s^2 - \left(M_q + \frac{Z_\alpha}{V_0} \right) s + \left(\frac{M_q Z_\alpha}{V_0} - M_\alpha \right)} \quad (4)$$

1.2 Lateral-Directional Dynamics

Due to the elimination of the rudder, the lateral control input is restricted strictly to the ailerons ($\Delta\delta_a$).

State-Space Representation:

$$x_{lat} = \begin{bmatrix} \Delta v \\ p \\ r \\ \Delta\phi \end{bmatrix}, \quad u_{lat} = [\Delta\delta_a] \quad (5)$$

$$\dot{x}_{lat} = A_{lat}x_{lat} + B_{lat,a}u_{lat} \quad (6)$$

$$A_{lat} = \begin{bmatrix} Y_v & Y_p + W_0 & Y_r - U_0 & g \cos \theta_0 \\ L_v & L_p & L_r & 0 \\ N_v & N_p & N_r & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad B_{lat,a} = \begin{bmatrix} Y_{\delta_a} \\ L_{\delta_a} \\ N_{\delta_a} \\ 0 \end{bmatrix} \quad (7)$$

Note: The N_{δ_a} term represents the adverse yaw induced by aileron deflection. Without a rudder, this disturbance is naturally damped by the aircraft's directional static stability (N_v or $C_{n\beta}$).

Roll Transfer Function: Assuming roll-dominated motion, the simplified aileron-to-roll-angle transfer function is:

$$\frac{\Phi(s)}{\Delta A(s)} = \frac{L_{\delta_a}}{s(s - L_p)} \quad (8)$$

2 Cascaded Control Architecture

To handle the integrators in the transfer functions and manage both energy and attitude robustly, the architecture employs a successive (cascaded) loop strategy.

2.1 Longitudinal Architecture (Energy and Pitch)

- **Airspeed Controller (Outer Loop 1):** A PI controller calculates the error between the reference airspeed (V_{ref}) and the measured pitot speed (V). The output directly dictates the **thrust command** ($\Delta\delta_t$).
- **Altitude Controller (Outer Loop 2):** A PID controller compares the reference altitude (h_{ref}) with the current altitude (h). The output yields the **reference pitch angle** (θ_{ref}).
- **Pitch Attitude Controller (Inner Loop 1):** A proportional (P) or PI controller takes the pitch error ($\theta_{ref} - \theta$) and generates a **reference pitch rate** (q_{ref}).
- **Pitch Rate Controller (Inner Loop 2):** A fast PID controller takes the rate error ($q_{ref} - q$) and generates the final **elevator command** ($\Delta\delta_e$).

2.2 Lateral Architecture (Navigation and Roll)

- **Heading/Waypoint Controller (Outer Loop):** The guidance logic calculates the desired heading (ψ_{ref}) towards the next waypoint. The error between ψ_{ref} and the current heading (ψ) generates the **reference roll angle** (ϕ_{ref}).
- **Roll Attitude Controller (Inner Loop 1):** A P or PI controller computes the error ($\phi_{ref} - \phi$) and generates a **reference roll rate** (p_{ref}).

- **Roll Rate Controller (Inner Loop 2):** A fast PID controller computes the rate error ($p_{ref} - p$) and outputs the final **aileron command** ($\Delta\delta_a$).

3 Flight Phases and State Machine

To perform a complete mission in a simulation environment (e.g., MATLAB/Simulink), a Stateflow or discrete logic block manages the control references across different flight phases:

1. Takeoff:

- Thrust (δ_t) is saturated to maximum (e.g., 100%).
- Pitch reference (θ_{ref}) is held at 0° until rotation speed (V_{rotate}) is achieved.
- At rotation, θ_{ref} transitions to a positive climb gradient (e.g., $+10^\circ$).

2. Cruise and Waypoint Navigation:

- Once a safe altitude is reached, the Airspeed and Altitude outer loops are engaged.
- The Heading controller is activated, processing GPS coordinates to steer the UAV through predefined waypoints.

3. Landing:

- The UAV intercepts the glide slope. h_{ref} ramps down towards ground level.
- Airspeed reference (V_{ref}) is reduced to the approach speed, causing the thrust controller to spool down the motor.
- Near touchdown, a flare maneuver is triggered by adding a slight positive bias to θ_{ref} to arrest the descent rate.

4 Conclusion

This integrated GNC architecture establishes a robust foundation for fixed-wing UAV simulation and autonomous flight. By decoupling the outer guidance loops (Altitude, Airspeed, Heading) from the inner stabilization loops (Pitch Rate, Roll Rate), the system simplifies the tuning process. The elimination of the rudder streamlines control allocation, relying entirely on bank-to-turn mechanics. The subsequent step involves estimating aerodynamic derivatives via CFD or XFLR5 analysis to finalize the numerical PID gains.